Question:

What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?

Answer Choices:

A. 2.48 × 10^-8 cm

B. 5.92 × 10^7 cm/sec

C. 8.66 × 10^-6 cm

D. 3.14 × 10^-5 cm

E. 9.81 × 10^-4 cm

F. 5.27 × 10^-9 m

G. 1.23 × 10^-7 cm

H. 4.14 × 10^-10 m

I. 6.625 × 10^-27 erg sec

J. 0.512 × 10^6eV

Reasoning + Answer:

<think>
reasoning process here
de Broglie wavelength is given by the formula \(\lambda = \frac{h}{p}\) where \(h\) is the Planck's constant and \(p\) is the momentum of the electron.
The kinetic energy of electron is 1 eV which can be converted to Joules as 1 eV = 1.6 ÃĹ 10^-19 J.
The momentum \(p\) can be found using the formula \(KE = \frac{1}{2} m v^2\) where \(KE\) is the kinetic energy, \(m\) is the mass of the electron and \(v\) is the velocity of the electron.
To calculate the velocity, take KE to be 1.6 ÃĹ 10^-19 J and the mass of the electron to be 9.1 ÃĹ 10^-31 kg, this will be used in the kinetic energy formula.
Using \(KE = \frac{1}{2} m v^2\), velocity \(v\) can be calculated.
Now, de Broglie wavelength can be found by \(\lambda = \frac{h}{p}\), we already have \(p = mv\).
Using \(h = 6.625 ÃĹ 10^{-34} J s\), and after substitution and simplification, the result will be in m and it will be converted to cm as the other options are in cm.
</think>
<answer>
6.625 ÃĹ 10^-9 m
</answer>